DDI-Net 3.0

Explainable drug interaction intelligence

# Multi-Drug Interaction Research Report

6 drugs · 15 pairs

DRUGS ANALYSED

1 **Warfarin**
DB00682 · approved

2 **Fluconazole**
DB00196 · approved, investigational

3 **Simvastatin**
DB00641 · approved

4 **Clarithromycin**
DB01211 · approved

5 **Acetylsalicylic acid**
DB00945 · approved, vet_approved

6 **Metformin**
DB00331 · approved

| | |
|---|---|
| Report ID | DDI-20261015-201851-D70B8C24 |
| Generated | 15 October 2026, 20:18:51 (UTC, UTC+00:00) |
| Generated (UTC) | 2026-10-15T20:18:51.040Z |
| Model | BIO-GINE M4 — frozen checkpoint bd45f84e3c1b2c33 per the site's frozen data; the multi-drug response itself does not name it |
| API report version | 3.0 |
| Analysis produced by the service | 2026-10-15T20:18:50+00:00 |
| Response received by the browser | 2026-10-15T20:18:50.599Z |
| Service | http://127.0.0.1:8765 |
| Language | English |

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RESEARCH USE ONLY

DDI-Net is a research prototype — not a medical device and not clinical decision support. The absence of a documented interaction is not evidence of safety. The BIO-GINE score is a research signal of resemblance to documented pairs, not a probability of clinical harm.

# 1 Executive summary

- Drugs analysed: 6; unique pairs: 15. Each pair was examined on its own.

Pairs by kind of documentation found:

| | | |
|---|---|---|
| CI | Contraindication in a label | 0 |
| L | Named in a label | 0 |
| ≠ | Sources conflict | 0 |
| DB | Documented in a database | 9 |
| M | Model signal only | 3 |
| – | No documentation found — does not mean “safe” | 3 |

- Pairs flagged by the service for a closer look: 12.
- Pairs whose own record the model was trained on (score not counted): 1.

## Clinical outcomes (as the sources state them)

| PAIR | INTERACTION STATUS (OUTCOME ENGINE) | CLINICAL OUTCOMES (AS THE SOURCES STATE THEM) |
|---|---|---|
| Warfarin + Fluconazole | evidence, outcome unclear | Increased drug exposure |
| Warfarin + Simvastatin | documented interaction | Bleeding and blood clotting |
| Warfarin + Clarithromycin | evidence, outcome unclear | Increased drug exposure |
| Warfarin + Acetylsalicylic acid | documented interaction | Bleeding and blood clotting |
| Warfarin + Metformin | source unavailable | no documented outcome |
| Fluconazole + Simvastatin | evidence, outcome unclear | Increased drug exposure |
| Fluconazole + Clarithromycin | evidence, outcome unclear | Increased drug exposure |
| Fluconazole + Acetylsalicylic acid | evidence, outcome unclear | Increased drug exposure |
| Fluconazole + Metformin | source unavailable | no documented outcome |
| Simvastatin + Clarithromycin | evidence, outcome unclear | Increased drug exposure |
| Simvastatin + Acetylsalicylic acid | source unavailable | no documented outcome |
| Simvastatin + Metformin | source unavailable | no documented outcome |
| Clarithromycin + Acetylsalicylic acid | source unavailable | no documented outcome |
| Clarithromycin + Metformin | source unavailable | no documented outcome |
| Acetylsalicylic acid + Metformin | documented interaction | Blood glucose |

### Documented outcome categories shared by several pairs

- Bleeding and blood clotting: Warfarin + Simvastatin; Warfarin + Acetylsalicylic acid
- Increased drug exposure: Warfarin + Fluconazole; Warfarin + Clarithromycin; Fluconazole + Simvastatin; Fluconazole + Clarithromycin; Fluconazole + Acetylsalicylic acid; Simvastatin + Clarithromycin

This is an overlap of categories in pairwise data. It does not mean the effects add up, and says nothing about three or more drugs together.

> There is no overall score for the set, and none is computed: pairwise findings cannot be added up into a conclusion about several medicines together.

> Each pair is examined on its own. Pairwise findings say nothing about how three or more drugs behave together: no such model has been built or validated.

### Uncertainty and evidence gaps

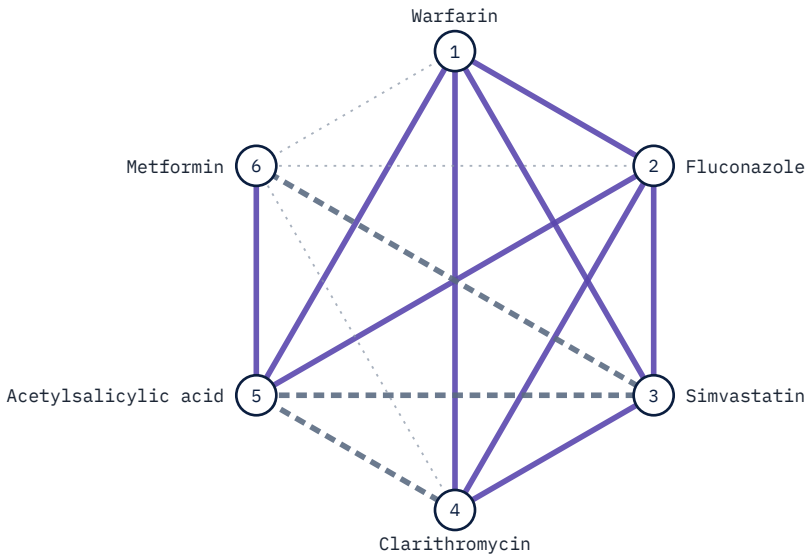
- Source “FDA labels (openFDA)”: unavailable. Its data is not part of this report.
- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.
- No source gives a severity, and none is assigned here.
- Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account.

## 2 Overview of the drugs

| # | INTERNATIONAL NAME (INN) | RUSSIAN NAME | DRUGBANK ID | DRUGBANK GROUPS | TYPE |
|---|---|---|---|---|---|
| 1 | warfarin | варфарин (transliteration) | DB00682 | approved | small molecule |
| 2 | fluconazole | флуконазол (transliteration) | DB00196 | approved, investigational | small molecule |
| 3 | simvastatin | симвастатин (transliteration) | DB00641 | approved | small molecule |
| 4 | clarithromycin | кларитромицин (-transliteration) | DB01211 | approved | small molecule |
| 5 | acetylsalicylic acid | ацетилсалициловая кислота (transliteration) | DB00945 | approved, vet_approved | small molecule |
| 6 | metformin | метформин (transliteration) | DB00331 | approved | small molecule |

Identifiers: DrugBank. INN: DrugCentral, joined on InChIKey (no value means no match). Russian names: rule-based transliteration — an interface label, not sourced data.

## 3 Pairwise map




Lines: solid — database or label record; dashed — model signal only; dotted — no documentation found. A thick line marks a pair flagged for a closer look.

### Pair matrix

Each cell is the kind of documentation found for the pair (row × column); the number is the calibrated model score. Colour encodes the kind of source, not danger. “n/c” — not counted: the model was trained on this pair's record.




**CI** Contraindication in a label **L** Named in a label **≠** Sources conflict **DB** Documented in a database **M** Model signal only **–** No documentation found

## 4 All pairs: evidence and model

For each pair the service reports the number of mechanism hypotheses; the hypotheses, proteins and pathways themselves are in the per-pair report (Analyzer). “Labels (excerpts)”: the number of FDA labels naming the other drug of the pair, and in brackets the distinct excerpts from them; several excerpts of one label are one source.

| PAIR | WHAT WAS FOUND | DRUGBANK | LABELS (EXCERPTS) | HYP. | MODEL | WHY LOOK CLOSER |
|---|---|---|---|---|---|---|
| Warfarin + Fluconazole<br>1–2 · DB00682 · DB00196 | Documented in a database | 1 | 0 (0) | 3 | 0.81<br>S2 | database record; pharmacokinetic mechanism hypothesis |
| Warfarin + Simvastatin<br>1–3 · DB00682 · DB00641 | Documented in a database | 1 | 0 (0) | 4 | 0.48<br>S2 | database record; pharmacokinetic mechanism hypothesis |
| Warfarin + Clarithromycin<br>1–4 · DB00682 · DB01211 | Documented in a database | 1 | 0 (0) | 2 | 0.49<br>S3 | database record; pharmacokinetic mechanism hypothesis |
| Warfarin + Acetylsalicylic acid<br>1–5 · DB00682 · DB00945 | Documented in a database | 1 | 0 (0) | 2 | 0.76<br>S3 | database record; pharmacokinetic mechanism hypothesis |
| Fluconazole + Simvastatin<br>2–3 · DB00196 · DB00641 | Documented in a database | 1 | 0 (0) | 3 | not counted<br>S1 | database record; pharmacokinetic mechanism hypothesis |
| Fluconazole + Clarithromycin<br>2–4 · DB00196 · DB01211 | Documented in a database | 1 | 0 (0) | 3 | 0.91<br>S2 | database record; pharmacokinetic mechanism hypothesis |
| Fluconazole + Acetylsalicylic acid<br>2–5 · DB00196 · DB00945 | Documented in a database | 1 | 0 (0) | 2 | 0.45<br>S2 | database record; pharmacokinetic mechanism hypothesis |
| Simvastatin + Clarithromycin<br>3–4 · DB00641 · DB01211 | Documented in a database | 1 | 0 (0) | 5 | 0.84<br>S2 | database record; pharmacokinetic mechanism hypothesis |
| Acetylsalicylic acid + Metformin<br>5–6 · DB00945 · DB00331 | Documented in a database | 1 | 0 (0) | 0 | 0.54<br>S2 | database record |

| PAIR | WHAT WAS FOUND | DRUGBANK | LABELS (EXCERPTS) | HYP. | MODEL | WHY LOOK CLOSER |
|---|---|---|---|---|---|---|
| Simvastatin + Acetylsalicylic acid<br>3–5 · DB00641 · DB00945 | Model signal only | 0 | 0 (0) | 4 | 0.55<br>S2 | model signal without documentation; pharmacokinetic mechanism hypothesis |
| Simvastatin + Metformin<br>3–6 · DB00641 · DB00331 | Model signal only | 0 | 0 (0) | 0 | 0.51<br>S1 | model signal without documentation |
| Clarithromycin + Acetylsalicylic acid<br>4–5 · DB01211 · DB00945 | Model signal only | 0 | 0 (0) | 2 | 0.51<br>S3 | model signal without documentation; pharmacokinetic mechanism hypothesis |
| Warfarin + Metformin<br>1–6 · DB00682 · DB00331 | No documentation found<br>does not mean “safe” | 0 | 0 (0) | 0 | 0.44<br>S2 | — |
| Fluconazole + Metformin<br>2–6 · DB00196 · DB00331 | No documentation found<br>does not mean “safe” | 0 | 0 (0) | 0 | 0.29<br>S1 | — |
| Clarithromycin + Metformin<br>4–6 · DB01211 · DB00331 | No documentation found<br>does not mean “safe” | 0 | 0 (0) | 0 | 0.50<br>S2 | — |

## 5 Coverage and uncertainty

- The regime says whether the pair's drugs were in the model's training set: S1 — both (not evaluated), S2 — one, S3 — neither.
- For each pair the service reports the number of mechanism hypotheses; the hypotheses, proteins and pathways themselves are in the per-pair report (Analyzer).
- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.

### Pairs whose score is not counted

- Fluconazole + Simvastatin (S1)

### Sources checked

| SOURCE | STATUS | DETAIL |
|---|---|---|
| DrugBank (TDC export, local) | checked | `Local DrugBank export (TDC).` |
| FDA labels (openFDA) | unavailable | `openfda_label: recently unreachable; retrying after a cooldown` |

## 6 Sources and reproducibility

| | |
|---|---|
| Endpoint (multi) | `http://127.0.0.1:8765/api/v3/multi` |
| Request body (multi) | `{"drugs":["DB00682","DB00196","DB00641","DB01211","DB00945","DB00331"],"include_labels":true}` |
| Response received by the browser (multi) | 2026-10-15T20:18:50.599Z |
| Report generated by the service (generated_at) | 2026-10-15T20:18:50+00:00 |
| API report version | 3.0 |
| Model | BIO-GINE M4 — frozen checkpoint bd45f84e3c1b2c33 per the site's frozen data; the multi-drug response itself does not name it |
| Frozen research tag | `v2-final-github-safe-2026-09-03 · 92c481eeaba8faff991ced850e1c4de418ea31b0` |
| Site | http://127.0.0.1:4173 |
| Site build commit | not recorded |
| PDF generator | DDI-Net web, in-browser export (@react-pdf/renderer) |
| Generated | 15 October 2026, 20:18:51 (UTC, UTC+00:00) |

| | |
|---|---|
| Generated (UTC) | 2026-10-15T20:18:51.040Z |
| Language | English |
| SHA-256 of the response data | d70b8c24bf3c949dc72490221abefa2f07d5376060dd4cebd085222a7f67d590<br>Hash of the service response(s) as canonical JSON (keys sorted, serialised with JSON.stringify). The same hash is in the companion JSON file, available from the "JSON" button next to "Export PDF". |

# 7 Limitations

## Core limitations

- DDI-Net is a research tool, not clinical decision support and not a medical device. Decisions about medicines belong to a doctor or pharmacist.
- Unlabelled pairs are not confirmed safe. A missing record means only that the sources checked do not contain one.
- BIO-GINE scores are not calibrated probabilities of clinical harm: temperature scaling was fitted on validation pairs at 50% prevalence against randomly drawn unlabelled pairs.
- The drug order in the DrugBank (TDC) export was verified by measurement for only two interaction types; for the others it is either meaningless (symmetric types) or unverified, so directional claims ("A raises the level of B") are not made.
- External medical evidence (FDA labels, RxNorm) may be incomplete or unavailable at query time; its status is listed under "Sources checked".
- Pairwise results say nothing about three or more drugs taken together; no such model exists here, and no overall score is computed.

> RESEARCH USE ONLY
> This is a research prototype, not a medical device. It does not replace a doctor or pharmacist and cannot say whether a combination is safe for a particular person.

# 8 Glossary

Plain explanations of the terms used in this report.

| | |
|---|---|
| **Source evidence vs model prediction** | Source evidence is text that an official label or a database actually contains about these two drugs. A model prediction is a number BIO-GINE computes from patterns in the data it was trained on. The first records what is known; the second is a research guess. |
| **FDA label (SPL)** | A medicine's official prescribing information in the United States, published by the FDA as Structured Product Labeling. One label is one document with its own set ID, divided into sections such as Warnings or Drug Interactions. |
| **Document and excerpt** | An excerpt is one sentence quoted from a document. Several excerpts of one label are still one source and do not confirm each other. When a label repeats a sentence in two sections, it is shown once. |
| **DrugBank (TDC export)** | A large database of drugs and known interactions; DDI-Net uses the public copy distributed by Therapeutics Data Commons. A record states only an interaction type, in a standard sentence, with no severity and no note of what it is based on. |
| **Direction** | Which drug affects which. For most interaction types the drug order in the DrugBank export is unreliable, so the report names no direction from DrugBank. Label sentences are quoted as written; no direction is inferred from the mere fact that one drug's label names the other. |
| **Cytochrome P450 (CYP450)** | A family of enzymes, mostly in the liver (for example CYP2C9, CYP3A4), that break down many medicines. If one drug slows an enzyme that breaks down another, the second may stay in the body longer. |
| **Substrate, inhibitor, inducer** | A substrate is a drug a protein processes. An inhibitor slows the protein down. An inducer makes the body produce more of it. |

| | |
|---|---|
| **Mechanism hypothesis** | A possible biological explanation assembled from what databases record about each drug separately (for example: drug X inhibits an enzyme that processes drug Y). It was not tested for this pair and says nothing about the size of any effect. |
| **Reactome pathway** | A curated chain of reactions in the cell. Two drugs sharing a pathway is background information, not a cause of interaction. |
| **Raw and calibrated score** | The raw score is the model's direct output between 0 and 1; the calibrated score is the same output rescaled on balanced validation pairs. Both put pairs in the same order. Neither is a probability of harm. |
| **Regimes S1, S2, S3** | Whether the model saw the drugs during training. S1: both drugs were in training (quality not measured). S2: one drug was not in training. S3: neither was — the hardest test. |
| **Held-out test; "pooled"** | Pairs set aside in advance and never used for training; quality is measured on them. "Pooled" means S2 and S3 test pairs measured together (84,690 pairs, 7,758 of them S3). |
| **AUROC** | The chance that the model ranks a randomly chosen documented pair above a randomly chosen pair from the comparison set (random pairs without a record). 0.5 is guessing; 1 is a perfect ranking. |
| **AUPRC** | Area under the precision–recall curve: how well the model finds documented pairs without raising many false alarms, over all thresholds at once. Half of the test pairs are documented, so guessing gives about 0.5; 1 is perfect. |
| **± (spread between runs)** | The model was trained 5 times, each from a different random starting point (seed). "±" is the standard deviation of the result between those runs. It is not an error bar for a single pair. |

# A Appendix A. Machine-readable summary (JSON)

Key fields as JSON for automated checking. The full service responses are in the companion JSON file (same SHA-256).


```
{
  "schema": "ddinet.pdf-summary/1",
  "report_id": "DDI-20261015-201851-D70B8C24",
  "generated_at": "2026-10-15T20:18:51+00:00",
  "generated_at_utc": "2026-10-15T20:18:51.040Z",
  "time_zone": "UTC",
  "language": "en",
  "payload_sha256": "d70b8c24bf3c949dc72490221abefa2f07d5376060dd4cebd085222a7f67d590",
  "inputs": [
    {"endpoint":"http://127.0.0.1:8765/api/v3/multi","request":{"drugs":["DB00682","DB00196","DB00641","DB01211","DB00945",
      "DB00331"],"include_labels":true},"received_at":"2026-10-15T20:18:50.599Z"}
  ],
  "server_generated_at": "2026-10-15T20:18:50+00:00",
  "report_version": "3.0",
  "drugs": [
    {"id":"DB00682","inn":"warfarin","name_ru":"варфарин"},
    {"id":"DB00196","inn":"fluconazole","name_ru":"флуконазол"},
    {"id":"DB00641","inn":"simvastatin","name_ru":"симвастатин"},
    {"id":"DB01211","inn":"clarithromycin","name_ru":"кларитромицин"},
    {"id":"DB00945","inn":"acetylsalicylic acid","name_ru":"ацетилсалициловая кислота"},
    {"id":"DB00331","inn":"metformin","name_ru":"метформин"}
  ],
  "pairs": [
    {"a":"DB00682","b":"DB00196","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":3,
      "score":0.8108211344965719,"regime":"S2","label_seen":false,"review":true,"interaction_status":"documented_outcome_unclear",
      "outcome_categories":["exposure_increase"]},
    {"a":"DB00682","b":"DB00641","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":4,
      "score":0.48439914908419174,"regime":"S2","label_seen":false,"review":true,"interaction_status":"documented_interaction",
      "outcome_categories":["bleeding_coagulation"]},
    {"a":"DB00682","b":"DB01211","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":2,
      "score":0.4894444855664803,"regime":"S3","label_seen":false,"review":true,"interaction_status":"documented_outcome_unclear",
      "outcome_categories":["exposure_increase"]},
    {"a":"DB00682","b":"DB00945","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":2,
      "score":0.7641836505184048,"regime":"S3","label_seen":false,"review":true,"interaction_status":"documented_interaction",
      "outcome_categories":["bleeding_coagulation"]},
    {"a":"DB00682","b":"DB00331","status":"no_documented_evidence_found","db":0,"label_documents":0,"label_excerpts":0,"hypotheses":
      0,"score":0.44394229455360606,"regime":"S2","label_seen":false,"review":false,"interaction_status":"source_unavailable",
      "outcome_categories":[]},
    {"a":"DB00196","b":"DB00641","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":3,
      "score":null,"regime":"S1","label_seen":true,"review":true,"interaction_status":"documented_outcome_unclear",
      "outcome_categories":["exposure_increase"]},
    {"a":"DB00196","b":"DB01211","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":3,
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      "outcome_categories":["exposure_increase"]},
    {"a":"DB00196","b":"DB00945","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":2,
      "score":0.44546109802252076,"regime":"S2","label_seen":false,"review":true,"interaction_status":"documented_outcome_unclear",
```

```
      "outcome_categories":["exposure_increase"]},
    {"a":"DB00196","b":"DB00331","status":"no_documented_evidence_found","db":0,"label_documents":0,"label_excerpts":0,"hypotheses":
      0,"score":0.29359325198856107,"regime":"S1","label_seen":false,"review":false,"interaction_status":"source_unavailable",
      "outcome_categories":[]},
    {"a":"DB00641","b":"DB01211","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":5,
      "score":0.8411772377234219,"regime":"S2","label_seen":false,"review":true,"interaction_status":"documented_outcome_unclear",
      "outcome_categories":["exposure_increase"]},
    {"a":"DB00641","b":"DB00945","status":"model_signal_only","db":0,"label_documents":0,"label_excerpts":0,"hypotheses":4,"score":
      0.5459016119192227,"regime":"S2","label_seen":false,"review":true,"interaction_status":"source_unavailable",
      "outcome_categories":[]},
    {"a":"DB00641","b":"DB00331","status":"model_signal_only","db":0,"label_documents":0,"label_excerpts":0,"hypotheses":0,"score":
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      "outcome_categories":[]},
    {"a":"DB01211","b":"DB00945","status":"model_signal_only","db":0,"label_documents":0,"label_excerpts":0,"hypotheses":2,"score":
      0.5057913508710338,"regime":"S3","label_seen":false,"review":true,"interaction_status":"source_unavailable",
      "outcome_categories":[]},
    {"a":"DB01211","b":"DB00331","status":"no_documented_evidence_found","db":0,"label_documents":0,"label_excerpts":0,"hypotheses":
      0,"score":0.4965563162607425,"regime":"S2","label_seen":false,"review":false,"interaction_status":"source_unavailable",
      "outcome_categories":[]},
    {"a":"DB00945","b":"DB00331","status":"documented_no_severity","db":1,"label_documents":0,"label_excerpts":0,"hypotheses":0,
      "score":0.5444521802812233,"regime":"S2","label_seen":false,"review":true,"interaction_status":"documented_interaction",
      "outcome_categories":["glucose"]}
  ],
  "sources_checked": [
    {"source":"drugbank_tdc","status":"ok"},
    {"source":"openfda_label","status":"unavailable"}
  ]
}
```